

Advanced Econometrics – Lab 02 – Mock test

Exercise 1

The exercise comes from the book *Principles of Econometrics*. The data file *cocaine.csv* describes variables related to sales of cocaine powder in northeastern California over the period 1984–1991. The data is a subset of those used in the study Caulkins, J.P. i R. Padman (1993), "Quantity Discounts and Quality Premia for Illicit Drugs", *Journal of the American Statistical Association*, 88, 748–757. Explanatory variables:

- PRICE - price per gram in dollars for a cocaine sale,
- QUANT - number of grams of cocaine in a given sale,
- QUAL - quality of the cocaine expressed as percentage purity,
- TREND - a time variable with $1984 = 1$ up to $1991 = 8$.

Let's consider the regression model of the form

$$PRICE = \beta_1 + \beta_2 QUANT + \beta_3 QUAL + \beta_4 TREND + \varepsilon. \quad (1)$$

Call:

```
lm(formula = price ~ quant + qual + trend, data = dane)
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Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	90.84669	8.58025	?????	??????
quant	-0.05997	0.01018	-5.892	2.85e-07 ***
qual	0.11621	0.20326	?????	??????
trend	-2.35458	1.38612	-1.699	0.0954 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 20.06 on 52 degrees of freedom

Multiple R-squared: 0.5097, Adjusted R-squared: ?????

F-statistic: 18.02 on 3 and 52 DF, p-value: 3.806e-08

- a) (10%) What signs of β_2, β_3 and β_4 should we expect? Explain. 
- b) (5%) How many observations were used to estimate the model? Explain. 
- c) (10%) Interpret the value of the coefficient of determination. 
- d) (5%) What is the value of the adjusted R^2 statistic? 
- e) (5%) Which variables are statistically significant at the level of $\alpha = 5\%$? Explain. 
- f) (5%) Are all variables jointly statistically significant at $\alpha = 5\%$? 
- g) (10%) Interpret the estimates of the parameters. 